## Problem 1

Heun method (2nd order Runge-Kutta)

$$2\frac{d^2y}{dx^2} + 3x^2y\frac{dy}{dx} = 10x^2$$

@ $x=1$, $y=2$, $\frac{dy}{dx}=2$

$$\frac{dy}{dx} = f(y,x)\ ;$$

$$\therefore\ 2\frac{df}{dx} + 3x^2yf = 10x^2$$

$$\frac{df}{dx} = 5x^2 - \frac{3}{2}x^2yf = g(x,y)$$

@ $x_0=1$, $f(1,2)=2$ ($y=2$), $g(1,2)=-1$

Discretize the domain: $x^n = x_0 + n\Delta x$, $\Delta x = 0.1$

Discretize the ODE: $\frac{\tilde{y}^{n+1}-y^n}{\Delta x} = f(y^n,x^n) \rightarrow \tilde{y}^{n+1} = y^n + \Delta x\, f(y^n,x^n)$

$$\frac{f^{n+1}-f^n}{\Delta x} = \frac{f(y^n,x^n)+g(\tilde{y}^{n+1},x^{n+1})}{2}$$

$$f^{n+1} = f^n + \frac{\Delta x}{2}g(\tilde{y}^{n+1},x^{n+1})$$

$$f(y^{n+1},x^{n+1}) = f(y^n,x^n) + \frac{\Delta x}{2}\left[5(x^{n+1})^2 - \frac{3}{2}(x^{n+1})^2\tilde{y}^{n+1}f(\tilde{y}^{n+1},x^n)\right]$$

$f^{n-1} = f(y^{n-1},x^{n-1})$ ; $f(x_0,y_0)=2$

$g$ ; $g(x_0,y_0)=-1$

| $n$ | $x^n = x_0 + n\Delta x$ | $\tilde{y}^n = y^{n-1} + \Delta x f^{n-1}$ | $\tilde{f}^n = f^{n-1} + \Delta x\, g(\tilde{y}^{n-1},x^{n-1})$ |
|---|---|---|---|
| 0 | $x^0 = x_0 = 1$ | — | — |
| 1 | $x^1 = 1+(0.1) = 1.1$ | $\tilde{y}^1 = y^0 + \Delta x f^0 = 2+(0.1)(2) = 2.2$ | $\tilde{f}^1 = f^0 + \Delta x\, g(\tilde{y}^0,x^0) = 2+(0.1)(-1) = 1.9$ |
| 2 | $x^2 = 1+2(0.1) = 1.2$ | $\tilde{y}^2 = y^1 + \Delta x f^1 = 2.195 + (0.1)(1.8732) = 2.3823$ | $\tilde{f}^2 = f^1 + \Delta x\, g(\tilde{y}^1,x^1) = 1.8732 + (0.1)(-1.4122) = 1.732$ |

$5(1.1)^2 - \frac{3}{2}(1.1)^3(2.195)(1.8732) = -1.4122$

| $f^n = f^{n-1} + \frac{\Delta x}{2}g(\tilde{y}^n,x^n)$ | $y^n = y^{n-1} + \frac{\Delta x}{2}(f^{n-1}+\tilde{f}^n)$ |
|---|---|
| — | — |
| $f^1 = f^0 + \frac{\Delta x}{2}g(\tilde{y}^1,x^1) = 2 + \frac{(0.1)}{2}\left(\underbrace{5(1.1)^2 - \frac{3(1.1)^2(2.2)(1.9)}{2}}_{5(1.1)^2-\frac{3}{2}(1.1)^2\tilde{y}^1\tilde{f}^1}\right) = 1.8732$ | $y^1 = y^0 + \frac{(0.1)}{2}(2+1.9) = 2.195$ |
| $f^2 = f^1 + \frac{\Delta x}{2}g(\tilde{y}^2,x^2) = 1.8732 + \frac{(0.1)}{2}\left(5(1.2)^2 - \frac{3(1.2)^2(2.3823)(1.73)}{2}\right) = -1.707$ | $y^2 = y^1 + \frac{(0.1)}{2}(f^1+\tilde{f}^2) = 2.195 + \frac{(0.1)}{2}(1.8732+1.732) = 2.3752$ |

Therefore; for $\Delta x = 0.1$ (w/ the given IC's) we get

$y^1 = 2.195$, $f^1 = \frac{dy^1}{dx} = 1.8732$ @ $x = 1.1$

$y^2 = 2.3752$, $f^2 = \frac{dy^2}{dx} = 1.7172$ @ $x = 1.2$

## Problem 2

2nd order Adams - Moulton method

$$2\frac{dy}{dx} + 3y^2 = 10 \quad , \quad y = 10 \text{ at } x = 1$$

$$\frac{dy}{dx} = 5 - \frac{3y^2}{2} = f(y,x) \quad ; \quad y_0(x_0 = 1) = 10$$

$$f(y_0, x_0) = 5 - 3\frac{(10^2)}{2} = -145$$

Implicit Adams - Moulton (Trapezoidal Method) :

$$y_{n+1} = y_n + \frac{\Delta x}{2}\left[f(y_n, x_n) + f(y_{n+1}, x_{n+1})\right]$$

@ n=0, $y_1 = y_0 + \frac{\Delta x}{2}\left[f(y_0, x_0) + f(y_1, x_1)\right]$

$$y_1 = 10 + \frac{(0.1)}{2}\left[-145 + 5 - 3\frac{y_1^2}{2}\right]$$

$$y_1 = 10 - 7 - 0.075\,y_1^2$$

$$0.075\,y_1^2 + y_1 - 3 = 0$$

Solving using Newton Rhapson : Let $h = 0.075\,y_1^2 + y_1 - 3$

$$h' = 0.15\,y_1 + 1$$

$$y_1^{(k+1)} = y_1^{(k)} - \frac{h(y_1^{(k)})}{h'(y_1^{(k)})}$$

@ k=0, $y_1^{(1)} = y_1^{(0)} - \frac{h(y_1^{(0)})}{h'(y_1^{(0)})}$

estimate using Euler → $y_1^{(0)} = y_0 + \Delta x\, f(y_0) = 10 + (0.1)(-145) = -4.5$

$h(y_1^{(0)}) = h(-4.5) = 0.075(-4.5)^2 + (-4.5) = -5.98125$

$h'(-4.5) = 0.15(-4.5) + 1 = 0.325$

$\therefore y_1^{(1)} = 13.9$

@ k=1,

$h(13.9) = 0.075(13.9)^2 + 13.9 - 3 = 25.39075$

$h'(13.9) = 0.15(13.9) + 1 = 3.085$

$y_1^{(2)} = 13 - \frac{25.39075}{3.085} = 5.67 \rightarrow y_1 \sim 5.67$

Using quadratic formula

$y_1 = \{-15.86, 2.5\}$ ✓ → still far off w/ two iterations but will just use this value

@ n=1 ; $y_2 = y_1 + \frac{\Delta x}{2}\left[f(y_1, x_1) + f(y_2, x_2)\right]$

$f(y_1, x_1) = 5 - \frac{3}{2}y_1^2 = 5 - \frac{3}{2}(5.67)^2 = -43.22$

$y_2 = 5.67 + \frac{\Delta x}{2}\left[-43.22 + 5 - \frac{3y_2^2}{2}\right]$

$y_2 = 5.67 + \frac{(0.1)}{2}\left[-43.22 + 5 - \frac{3y_2^2}{2}\right]$

$y = 3.759 + 0.075\,y_2^2$

$0.075\,y^2 + y - 3.759 = 0$

Solving using Newton Rhapson : Let $g = 0.075\,y_2^2 + y_2 - 3.759$

$$g' = 0.15\,y_2 + 1$$

$$y_2^{(k+1)} = y_2^{(k)} - \frac{g(y_2^{(k)})}{g'(y_2^{(k)})}$$

@ k=0, $y_2^{(1)} = y_2^{(0)} - \frac{g(y_2^{(0)})}{g'(y_2^{(0)})}$

d ~ d ✓

@ k=0,

$$y_2^{(1)} = y_2^{(0)} - \frac{g(y_2^{(0)})}{g'(y_2^{(0)})}$$

estimate using Euler → $y_2^{(0)} = y_2 + \Delta x f(y_1) = 5.47 + (0.1)(-43.22) = 1.348$

$g(y_2^{(0)}) = g(1.348) = 0.075(1.348)^2 + (-2.411) = -2.2747$

$g'(1.348) = 0.15(1.348) + 1 = 1.2022$

$\therefore\ y_2^{(1)} = 1.348 - \frac{(-2.2747)}{(1.2022)} = 3.24$

@ k=1,

$g(13.9) = 0.075(3.24)^2 + 3.24 - 3.759 = 0.26832$

$g'(13.9) = 0.15(3.24) + 1 = 1.486$

$y_2^{(2)} = 3.24 - \frac{0.26832}{1.486} = 3.0594 \rightarrow \quad y_2 \approx 3.06$

Using quadratic formula

$y_1 = \{-16.39, 3.0587\}$ ✓ → much better approximation

$\therefore$ For

| | | |
|---|---|---|
| $x_1 = 1.1,$ | $y_1 = 5.47$ | w/ $\Delta x = 0.1$ |
| $x_2 = 1.2,$ | $y_2 = 3.06$ | & given initial conditions |

## Problem 3

Steady state conduction in a flat plate

$L \Rightarrow$ plate thickness

$\dot{q} \Rightarrow$ heat generation

$$\frac{d}{dx}\left(k\frac{dT}{dx}\right) + \dot{q} = 0$$

$k = a + bT$

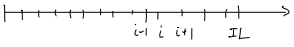
$\dot{q} = d + eT$

$$-k\frac{dT}{dx}\bigg|_{x=0} = h_H\left(T_H - T(x=0)\right)$$

$$-k\frac{dT}{dx}\bigg|_{x=L} = h_L\left(T(x=L) - T_c\right)$$

$-k_1\frac{(T_2 - T_1)}{\Delta x} = h_H(T_H - T_1)$ ($k_1 = a + bT_1$) $\Big\} F_{BC}^{(1)}(T_1, T_2)$

$-k_2\frac{(T_{IL} - T_{IL-1})}{\Delta x} = h_L(T_{IL} - T_c)$ $\Big\} F_{BC}^{(4)}(T_{IL}, T_{IL-1})$




Discretizing the equation:

$$\left[\frac{d}{dx}\left(k\frac{dT}{dx}\right) + \dot{q}\right]_i = \frac{k\frac{dT}{dx}\Big|_{i+\frac{1}{2}} - k\frac{dT}{dx}\Big|_{i-\frac{1}{2}}}{\Delta x} + q_i = 0$$

$$F(T_{i-1}, T_i, T_{i+1}) = \frac{\left(k_{i+\frac{1}{2}}\frac{T_{i+1} - T_i}{\Delta x} - \left(k_{i-\frac{1}{2}}\right)\frac{T_i - T_{i-1}}{\Delta x}\right)}{\Delta x} + d + eT_i = 0 \Big\} F(T_{i+1}, T_i, T_{i-1}) = 0,\ i = 2, 3, \ldots, IL-1$$

$k_{i+\frac{1}{2}} = \frac{k_i + k_{i+1}}{2} = \frac{(a+bT_i) + (a+bT_{i+1})}{2}$

$k_{i-\frac{1}{2}} = \frac{k_i + k_{i-1}}{2} = \frac{(a+bT_i) + (a+bT_{i-1})}{2}$

$$= \left(\frac{(a+bT_i)}{2(\Delta x)^2} + \frac{(a+bT_{i+1})}{2(\Delta x)^2}\right)(T_{i+1} - T_i) - \left(\frac{a+bT_i}{2(\Delta x)^2} + \frac{a+bT_{i-1}}{2(\Delta x)^2}\right)(T_i - T_{i-1}) + d + eT_i = 0$$

$$= \frac{(a+bT_i)T_{i+1}}{2(\Delta x)^2} + \frac{(a+bT_{i+1})T_{i+1}}{2(\Delta x)^2} - \frac{(a+bT_i)T_i}{2(\Delta x)^2} - \frac{(a+bT_{i+1})T_i}{2(\Delta x)^2} - \frac{(a+bT_i)T_i}{2(\Delta x)^2} - \frac{(a+bT_{i-1})T_i}{2(\Delta x)^2} + \frac{(a+bT_i)T_{i-1}}{2(\Delta x)^2} + \frac{(a+bT_{i-1})T_{i-1}}{2(\Delta x)^2} + d + eT_i = 0$$

$$\frac{\partial F}{\partial T_{i-1}} = -\frac{bT_i}{2(\Delta x)^2} + \frac{bT_{i-1}}{(\Delta x)^2} + \frac{a}{2(\Delta x)^2}$$

$$\frac{\partial F}{\partial T_i} = \frac{bT_{i+1}}{2(\Delta x)^2} - \frac{2bT_i}{(\Delta x)^2} - \frac{a}{(\Delta x)^2} - \frac{(a+bT_{i+1})}{2(\Delta x)^2} - \frac{(a+bT_{i-1})}{2(\Delta x)^2} + \frac{bT_{i-1}}{2(\Delta x)^2} + e = \left(e - \frac{a}{(\Delta x)^2}\right) + \left(-\frac{a}{2(\Delta x)^2} - \frac{T_{i+1}}{2(\Delta x)^2} + \frac{bT_{i+1}}{2(\Delta x)^2}\right) + \left(-\frac{2bT_i}{(\Delta x)^2}\right) + \left(\frac{bT_{i-1}}{2(\Delta x)^2} - \frac{a+bT_{i-1}}{2(\Delta x)^2}\right)$$

$$\frac{\partial F}{\partial T_{i+1}} = \frac{(a+bT_i)}{2(\Delta x)^2} + \frac{a}{2(\Delta x)^2} + \frac{bT_{i+1}}{(\Delta x)^2} - \frac{bT_i}{2(\Delta x)^2} = \frac{a}{2(\Delta x)^2} + \frac{bT_i}{2(\Delta x)^2} + \frac{a}{2(\Delta x)^2} + \frac{bT_{i+1}}{(\Delta x)^2} - \frac{bT_i}{2(\Delta x)^2} = \frac{a}{(\Delta x)^2} + \frac{bT_{i+1}}{(\Delta x)^2}$$

Boundary conditions ($(i) \rightarrow$ initial ; $(f) \rightarrow$ final)

$$-k\left.\frac{dT}{dx}\right|_{x=0} = h_H\left(T_H - T(x=0)\right)$$

$$-k_1\frac{(T_2 - T_1)}{\Delta x} = h_H(T_H - T_1)$$

$(a+bT_1)$

Initial

$$F_{BC}^{(i)} = \left(\frac{(a+bT_1)T_2 - (a+bT_1)T_1}{\Delta x}\right) + h_H T_H - h_H T_1 = 0$$

$$\frac{\partial F_{BC}^{(i)}}{\partial T_1} = \frac{bT_2}{\Delta x} - \frac{2bT_1}{\Delta x} - h_H - \frac{a}{\Delta x}$$

$$\frac{\partial F_{BC}^{(i)}}{\partial T_2} = \frac{(a+bT_1)}{\Delta x}$$

$$F(T_1, T_2) = (a+bT$$

Newton raphson

$$\begin{bmatrix} \frac{\partial F_1}{\partial T_1} & \frac{\partial F_2}{\partial T_2} \\ \cdots & \cdots \end{bmatrix}$$

$$-k\left.\frac{dT}{dx}\right|_{x=L} = h_L\left(T(x=L) - T_c\right)$$

$$-k_2\frac{(T_{IL} - T_{IL-1})}{\Delta x} = h_L(T_{IL} - T_c)$$

$(a+bT_{IL})$

$$F_{BC}^{(f)} = \left(\frac{(a+bT_{IL})T_{IL} - (a+bT_{IL})T_{IL-1}}{\Delta x}\right) + h_L(T_{IL} - T_c)$$

$$\frac{\partial F_{BC}^{(f)}}{\partial T_{IL}} = \frac{2bT_{IL}}{\Delta x} - \frac{bT_{IL-1}}{\Delta x} + h_L + \frac{a}{\Delta x}$$

$$\frac{\partial F_{BC}^{(f)}}{\partial T_{IL-1}} = -\frac{(a+bT_{IL})}{\Delta x}$$

Therefore to solve second order non-linear equation, we can construct the following $Ax=b$ matrix and use Thomas algorithm to find $\Delta T$ values.

$$\begin{bmatrix} \frac{\partial F_{BC}^{(i)}}{\partial T_1} & \frac{\partial F_{BC}^{(i)}}{\partial T_2} & & & \\ & \ddots & & & \\ & \frac{\partial F}{\partial T_{i-1}} & \frac{\partial F}{\partial T_i} & \frac{\partial F}{\partial T_{i+1}} & \\ & & \ddots & & \\ & & \frac{\partial F_{BC}^{(f)}}{\partial T_{IL-1}} & \frac{\partial F_{BC}^{(f)}}{\partial T_{IL}} & \end{bmatrix}_{\vec{T}_{old}} \begin{bmatrix} \\ \Delta\vec{T} \\ \\ \end{bmatrix} = - \begin{bmatrix} \\ \vec{F}(T_{old}) \\ \\ \end{bmatrix}$$

$$\vec{T}_{new} = \vec{T}_{old} + \Delta\vec{T}$$

$$\vec{T}_{old} = \vec{T}_{new} + \Delta\vec{T}$$

A constraint of $|\Delta\vec{T}| \geq \varepsilon \cdot |\vec{T}_{new}|$ is used, where $\varepsilon = 1e^{-5}$

Initial cond. of 1000 K are chosen

The above algorithm is coded in python (w/ the help of Chatgpt to debug the code) and the following plots are obtained

The decrease in temperature (from hot end of ~2000 K) to cold end of ~400 K makes sense since the rod is cooling down. This is also shown by the graphs.

-additionally, the grid spaces were varied from 10 to 150 to give the same results (grid independent)

